Metadata, cost and accuracy of official statistics served to AI agents: evidence from three MCP servers

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Abstract

AI agents increasingly reach official statistics through Model Context Protocol (MCP) servers rather than through portals or application programming interfaces. A value that reaches an agent without its source, status, uncertainty or release cannot be attributed, dated or interpreted correctly. This paper proposes a minimum metadata standard of fifteen elements for MCP servers that republish official statistics. It scores three public servers against it: the UNICEF SDMX data warehouse MCP, the UN System Data Commons (UN DC) and World Bank Data360. It then measures what answering a question costs on each. Three designs are used: a scripted pilot on two Nigerian series; a pre-registered benchmark of 34 randomly drawn series across 17 sectors and three countries (140 answers, 325 calls); and an agentic benchmark of 96 sessions across seven model and effort settings. The warehouse delivers 46–70% of the standard on every question it answers, against at most 21% for UN DC and 20–29% for Data360. Compliance is cheap, because the metadata travels in the data call, which accounts for 6–22% of characters on UN DC and the warehouse; searching for the series accounts for 67–94%. Fixed-effects models, with standard errors clustered at the level of sampling and wild cluster bootstrap inference, put numbers on these differences: within the same question, UN DC delivers 37.9 percentage points less of the standard than the warehouse, at 35.2% lower cost in characters. Across models and effort levels, every reported value for a year the warehouse holds (1,952 of 1,953) matched it, and all 1,953 appeared in the servers' own responses. Higher effort raised cost, not accuracy: Opus at high effort sent 36.5% more tokens than at low effort, against 7.3% between two identical runs. In agentic use, the tool definitions a server loads into every request cost more tokens than its data responses. For an official UNICEF MCP, the levers are a compact search, a short tool list, and one data call that carries the fifteen elements.

Keywords: official statistics; metadata; SDMX; Model Context Protocol; AI agents; data dissemination.

1 Introduction

When an analyst asks an AI assistant for Nigeria's under-five mortality rate, the number usually no longer comes from a portal, a PDF or a download. An agent calls a tool exposed by an MCP server, reads the response and writes an answer. Whatever the server leaves out of that response, the answer cannot contain: who produced the estimate, which release it belongs to, whether it was modelled or reported, and how uncertain it is. For official statistics, whose value rests on provenance and method as much as on the number, this is a new and largely unexamined channel.

*Working draft. Every number, table and figure is generated from the replication package that accompanies this paper; see the Reproducibility section.

This paper asks two questions. First, what metadata must a server return with each value for an agent to reproduce, attribute, date and interpret it, and how far do current servers meet that bar? Second, what does it cost, in characters and tokens, to answer a statistical question through each server, and does that cost depend on the metadata, on the server's design, or on the model that drives the agent?

We answer them for three public MCP servers that republish UNICEF's child-related indicators: the UNICEF SDMX data warehouse MCP (sdmx-mcp 1.2.214), the UN System Data Commons MCP, and the World Bank Data360 MCP. We propose a minimum standard of fifteen elements, each reusing an attribute already defined by SDMX (SDMX Sponsors), the SDG data structure definition (United Nations Statistics Division, 2019) or the UNICEF warehouse. We score the three servers against it, and run three benchmarks of increasing realism. Differences are tested with exact paired tests and estimated with linear fixed-effects models: question fixed effects for comparing servers, and question × server fixed effects for comparing models and effort levels. Standard errors are clustered by series and by question × server cell respectively, with wild cluster bootstrap inference because the clusters are few.

We find four things.

- **Coverage of the standard ranks the servers consistently.** The warehouse delivers 46–70% of the standard wherever it answers, against at most 21% for UN DC and 20–29% for Data360; within the same question, UN DC delivers 37.9 percentage points less than the warehouse and Data360 31.6 points less.
- **Compliance is not a cost problem.** The metadata travels in the data call, the cheap part of any answer; searching for the right series accounts for 67% of the warehouse's characters and 94% of UN DC's. UN DC is 35.2% cheaper than the warehouse on average: its typical search is somewhat cheaper, it needs no structure call, and its data call is smaller, partly because it carries less metadata. Even so, the warehouse's metadata-bearing data call is the smallest part of its own cost.
- **Accuracy did not depend on the model or the effort level.** Every value agents reported for a year the warehouse holds (1,952 of 1,953) matched it, and all 1,953 appeared in the servers' own responses. Higher effort bought more calls and tokens, not better answers: 36.5% more tokens sent for Opus and 12.6% for Sonnet, against a placebo difference of 7.3% between two identical runs.
- **The tool definitions a server loads cost more than its data.** In agentic use they are re-sent with every request and add 70,000–80,000 tokens per question on average; all data responses and reasoning together add 17,000–33,000.

Section 2 motivates the problem. Section 3 sets out the standard, the scoring and the cost measures, and Section 4 the design of the experiment, including the estimated models and how their standard errors are clustered. Section 5 describes the servers, samples and data recorded. Section 6 presents results, Section 7 draws policy implications and Section 8 concludes with limitations.

2 Motivation

An official statistic is a value plus its metadata. The same national under-five mortality rate can be a modelled UN IGME estimate with a 90 per cent uncertainty interval (UN IGME, 2025), or a direct estimate from one survey round. A coverage rate can come from the latest WHO/UNICEF revision or from the one before it. The international statistical system has spent two decades codifying how this information travels between agencies. The SDG data structure definition makes the unit, unit multiplier and observation status mandatory, and defines source detail, time detail, bounds and observation comments (United Nations Statistics Division, 2019). The Statistical Commission's guidelines on data flows and global data reporting set out how data and metadata should pass between national systems and custodian agencies (United Nations

Statistical Commission, 2018), and the CCSA principles ask for credit to the original source (Committee for the Coordination of Statistical Activities, 2005). None of this yet reaches the channel through which AI agents now read the data.

Republishing multiplies the risk of loss. A UNICEF value may reach an agent directly from the warehouse, or through a republisher that copied it from the SDG Global Database or from a World Bank collection. At each hop attributes can be dropped, codes re-mapped and releases fall out of date. Our earlier comparisons found concrete cases of each: UN DC attributes every UNICEF facet to "United Nations"; Data360 credits UNICEF on 65 of 117 matched series; and both still served Nigeria's pre-revision DTP3 coverage for 2021–2024 (61, 59, 67, 67), where the warehouse, on the WUENIC 2025 revision, has 50, 55, 63 and 63. An agent cannot detect any of these from the response alone.

Cost shapes what agents can use. Agents are billed and limited by tokens. A server that returns 30,000 characters where 3,000 would do makes every answer slower and dearer, crowds the model's context, and may push the agent to stop early or to skip a metadata call. A standard that made responses much longer would not be adopted; a server that is cheap only because it drops metadata is not a good server.

There is no standard yet. The MCP specification defines how a tool declares an output schema and returns structured content, but not what a statistical response should contain (Model Context Protocol, 2026). UNICEF is preparing an official MCP server for its data, and the UN system is building shared infrastructure through the UN System Data Commons under UN80 Workstream 16. Both need an agreed minimum and evidence on what it costs.

3 Methodology

3.1 A minimum metadata standard

An element enters the standard only if its absence caused a documented failure in our comparisons of the three servers against the warehouse, and each reuses an attribute that SDMX, the SDG data structure definition or the UNICEF warehouse already defines. Elements attach at three levels: R elements once per response, S elements once per series, and O elements per observation, only when non-empty. Twelve are required in every response; three (marked *) whenever the custodian publishes them (Table 1).

Seven carriage rules accompany the elements: (1) pass every element through and never drop one; (2) inline what interpretation needs and link the rest; (3) send header fields once per series and per-observation fields only where they vary; (4) send every coded field as code and label; (5) declare every default the server applied; (6) return values as published, without extra rounding or silent scaling; (7) declare the response schema through MCP's `outputSchema`.

3.2 Scoring

Each answer is scored element by element as *meets* (1 point), *partial* (0.5) or *missing* (0). Partial means incomplete, uneven across flows, or available only through a separate call. Elements the custodian does not publish for that series and country are not applicable. The *share of the standard delivered* is points divided by applicable elements. The automated scoring reproduces all ten hand-scored pilot paths element by element.

Table 1: The minimum metadata standard

ID	Element	What it must hold	Reuses
R1	Reproducible query	the canonical request that returns the same data	SDMX REST query
R2	Retrieval stamp	server name and version; retrieval time (UTC)	MCP serverInfo; ISO 8601
R3	Completeness statement	every default or filter applied; whether truncated	warehouse MCP appliedDefaults
S1	Identifiers	custodian code unchanged, server id, SDG series code	SDMX INDICATOR; SDG SERIES
S2	Name and definition	name, short definition, link to methodology	SDMX codelists; SDG metadata
S3	Unit	unit code and label, unit multiplier	SDMX UNIT_MEASURE, UNIT_MULT
S4	Breakdown	every dimension coded, explicit totals	SDMX dimensions, _T totals
S5	Custodian and provenance chain	estimating agency and each republisher	SDG custodian role; CCSA principle 4
S6	Release and currency	release, date, ingest date, current or replaced	SDMX dataflow version
S7	Licence and citation	licence and recommended citation	FAIR R1.1
O1	Source	survey round, administrative system or model	SDMX DATA_SOURCE; SDG SOURCE_DETAIL
O2	Reference period*	the period the value refers to, if different	SDG TIME_DETAIL; COVERAGE_TIME
O3	Nature and status	reported, adjusted, modelled, projected; provisional	SDMX OBS_STATUS; SDG NATURE
O4	Uncertainty*	lower and upper bounds, interval type and level	LOWER_BOUND, UPPER_BOUND
O5	Footnote*	the custodian's note, verbatim	SDG COMMENT_OBS; OBS_FOOTNOTE

* Required whenever the custodian publishes it.

3.3 Cost

Cost is measured in characters of every MCP response, counted exactly from the agents' session transcripts (oversized responses from the copy the client saved), and, in the agentic benchmark, in tokens: the usage the API reported for each request. Tokens sent are input plus cache-creation plus cache-read tokens, summed over requests; tokens received are output tokens, thinking included. Tokens sent are decomposed into the *baseline* (the first request, i.e. the client's own instructions and tool list, times the number of requests), *tool definitions* (the jump in context when the server's tools load, times later requests) and the *remainder* (responses and reasoning carried forward). A transcript repeats the usage of one API response on each of its content blocks, so usage is counted once per message identifier.

3.4 Frontier, accuracy and grounding

A way of answering is on the *efficiency frontier* if no other way delivers at least the same share of the standard for fewer characters. A value is *correct* if it is within the larger of 0.5% relative error and the rounding unit of the value reported (0.51 for integers, 0.051 otherwise) of the warehouse value for that year. A number is *grounded* if it appears in the server's responses in that session, verbatim or rounded to the precision reported.

4 Design of the experiment

The study combines controlled measurement with one experiment. The servers are three fixed systems, so comparisons between them are within-question contrasts, not effects of an intervention. The models and effort levels, by contrast, were crossed with every question and server, so their contrasts are experimental effects for this panel. The study asks four questions. Three designs answer them, and each design holds fixed what the others vary (Table 2).

Q1 Do the servers differ in the share of the standard they deliver, and is the ranking stable across sectors and countries?

Q2 Does delivering the standard make an answer materially more expensive?

Q3 What drives the cost of an answer: finding the series, its structure, or the data and metadata?

Q4 Do the model and its effort level change the cost or the accuracy of an answer on a given server?

Table 2: Benchmark designs

Design	Questions	Who chooses the calls	Measures	Purpose	Answers
Pilot	2 series, Nigeria; 13 paths	analyst, scripted	characters, share	build and calibrate the scoring; first frontier	Q1–Q3
Sector and country	34 series × up to 3 countries; 140 answers	fixed path per server, pre-registered	characters by component, share, discovery, agreement	test whether the pilot generalises	Q1–Q3
Model and effort	4 series, Nigeria × 3 servers × 8 runs; 96 sessions	the agent	tokens sent and received, calls, accuracy, grounding, stability	separate the server from the model	Q4

4.1 Units, factors and assignment

The unit of analysis is one *answer*: one question (a series, a country and the window 2000–2025) answered on one server. In the agentic design it is one *session*: one agent, one question, one server. Table 3 lists the factors and how their levels were assigned.

Table 3: Factors, levels and assignment

Factor	Levels	How assigned
Server	UNICEF SDMX warehouse, UN DC, Data360	crossed with every question, so each question is answered on every server that holds it (paired comparison)
Series	34 (sector design); 4 (agentic design)	sector design: one code drawn at random in each of 34 strata (17 UN DC sectors × lower and upper half of breakdown richness), seed 20261027; agentic design: purposive, the series all three servers hold with agreeing values
Country	Nigeria; Brazil and Somalia	Nigeria for every series; Brazil and Somalia for 12 series drawn at random
Answer path	one per server; or the agent's own	sector design: fixed in advance (search with the custodian's name, one prescribed retry, structure call, data call); agentic design: chosen by the agent
Model	Claude Haiku 5.5, Sonnet 5.5, Opus 5.5	crossed with every question and server
Effort	low, medium, high (Haiku: medium only)	crossed with Sonnet and Opus
Repeat	Opus at medium effort, twice	the whole 12-session block run again

4.2 Controls

Each control removes one alternative explanation for a difference between servers or settings.

- **Paired comparisons.** Every server answers the same questions, so differences in a series' difficulty, breakdowns or coverage cancel within a question.
- **Same window everywhere.** All requests cover 2000–2025. Data360's pilot calls, first made for longer windows, were re-run on that window; both runs are kept and the first is flagged.
- **Fixed discovery queries.** In the sector design, the search term is the custodian's indicator name, with one prescribed retry, so no server benefits from a query tuned to it.
- **Oracle fallback.** The data call always uses the target identifier, even when discovery fails. Retrieval cost is therefore measured on every question, and discovery failure is recorded separately rather than confounding it.
- **Isolation of the agent.** In the agentic design, each session uses one server, with no web search, files or prior knowledge, the same prompt, and a fixed answer form that asks for null where the server gave nothing. What an agent reports can then only come from the server.
- **A matched panel.** The four agentic series agree across the three servers, so differences in accuracy cannot come from the data themselves.
- **Exact measurement.** Characters are counted from the transcripts and tokens taken from the usage the API reported, never from what an agent says it did. Every call is counted, failures included.
- **Scoring fixed before use.** The automated scoring rules were validated against all ten hand-scored pilot paths before the sector benchmark was scored.
- **Pre-registration.** The sector design's question, sample, paths and measures were written into a protocol before the first call. The agentic design's panel, prompt and settings were fixed before its first session, but its hypotheses were not registered.

4.3 Outcomes

The primary outcomes are the share of the standard delivered (Q1, Q2), characters per answer (Q2, Q3), tokens sent and received per question (Q4) and the accuracy of reported values against the warehouse (Q4). Secondary outcomes are cost by component, discovery success, agreement and latest year against the warehouse, grounding of reported numbers and metadata, the number of metadata fields reported, stability across the repeat, and the determinism of server responses to identical calls.

4.4 Inference

Inference has two layers. This subsection reports exact, distribution-free tests, which need no model: two-sided sign tests for paired comparisons, and one-sided 95% upper bounds for outcomes with no observed failures. Section 4.5 then estimates average effects with fixed-effects models and cluster-robust inference.

- **Standard (Q1).** The warehouse delivered more of the standard than UN DC on 51 of 51 questions both answered ($p < 10^{-12}$), and more than Data360 on 22 of 22 ($p < 10^{-6}$).
- **Cost (Q2, Q3).** UN DC was cheaper than the warehouse on 43 of 58 questions ($p = 0.0003$), and Data360 on 22 of 24 ($p < 10^{-4}$); Data360's figure partly reflects failed searches (Section 6).
- **Effort (Q4).** Compared within each of the 12 question × server cells, high effort sent more tokens than low effort in 12 of 12 cells for Opus and 12 of 12 for Sonnet ($p = 0.0005$ each). The median increase was 33% for Opus and 6% for Sonnet. Running Opus at medium effort again changed a cell's tokens sent by a median of 2% and at most 22%. The direction of the effort effect is therefore clear; its size, from one run per setting, is approximate.
- **Accuracy (Q4).** No session reported a wrong value (0 of 96). Because values within a session are not independent, the honest bound is per session: the one-sided 95% upper bound on the share of sessions with any wrong value is 3.1%. Treating the 1,952 values as independent would give 0.15%, which overstates the precision.

The sign tests treat questions and cells as independent. Series from the same dataflow share server code paths, so these p-values are indicative; the models of Section 4.5 relax this assumption by clustering.

4.5 Estimating average effects

The sign tests say whether one server or setting is better in most matched comparisons; they do not say by how much on average. Average effects are estimated with two linear fixed-effects models, by ordinary least squares.

Servers (Q1–Q3). For answer j (server) to question i (series s in country c):

$$y_{ij} = \alpha_i + \beta_U \,\mathrm{UNDC}_j + \beta_D \,\mathrm{Data360}_j + \varepsilon_{ij}, \tag{1}$$

where α_i is a question fixed effect and the warehouse is the omitted server. β_U and β_D are the average differences from the warehouse *within the same question*, so series that are harder to find, more disaggregated or sparser in a country do not bias the comparison, and the unbalanced Data360 sample (24 of 58 questions) is handled without dropping the others. Two outcomes are modelled: the share of the standard in percentage points, on answered questions, and ln(characters), on all questions. Cost enters in logarithms because it is strongly right-skewed (from about 500 to 128,000 characters) and because servers plausibly scale cost multiplicatively; $100\,(e^{\beta} - 1)$ is then the percentage difference in the geometric mean.

Model and effort (Q4). For session k in cell c (a question × server pair) run with setting m:

$$\ln y_{cm} = \alpha_c + \sum_{m \neq \text{Sonnet, medium}} \gamma_m S_m + u_{cm}, \tag{2}$$

where α_c is a cell fixed effect and S_m indicates the setting, with Sonnet at medium effort omitted. The quantities of interest are contrasts: $\gamma_{\text{Opus, high}} - \gamma_{\text{Opus, low}}$ and $\gamma_{\text{Sonnet, high}} - \gamma_{\text{Sonnet, low}}$ for effort, $\gamma_{\text{Haiku, medium}}$ for model size, and $\gamma_{\text{Opus, medium, repeat}} - \gamma_{\text{Opus, medium}}$, a placebo whose true value is zero. Outcomes are ln tokens sent, ln tokens received and the number of MCP calls (in levels). Accuracy is not modelled: it does not vary (no wrong value in any session), so it is reported with the zero-event bound of Section 4.

How standard errors are clustered, and why. Clustering should follow how the data were generated: at the level at which units were sampled or treatments assigned, and wherever errors are correlated across observations (Abadie et al., 2023; MacKinnon et al., 2023).

- **Equation (1): cluster by series ($G = 34$).** Series, not questions or answers, were the unit of random sampling: one code was drawn per stratum, and countries were added for whole series. The answers for one series are also correlated across countries and servers. The discovery query is the same in every country, so search costs are nearly identical (the warehouse returned 5,787–5,789 characters for the same economic series in all three countries), and a series' metadata come from the same dataflow wherever it is asked. Clustering by question would treat the three countries of a series as independent: on these data it gives standard errors up to 47% smaller than clustering by series, and heteroskedasticity-robust errors are smaller still (by up to 86%). Clustering by server is not an option: the server is the treatment, and three clusters support no inference.
- **Equation (2): cluster by cell ($G = 12$).** The settings were crossed with every cell, so assignment is at the session level and the cell fixed effects absorb anything a cell's sessions share. Clustering is still needed because the effect of a setting can differ across cells—high effort may cost more on one server than another—and that heterogeneity makes the residuals of the sessions in a cell correlated. Inference about the average effect over cells must allow for it. Questions (4) and servers (3) are too few to cluster on.
- **Few clusters.** With 12 clusters, and even 34, the usual cluster-robust (CR1) variance with $t(G-1)$ critical values can over-reject (Cameron et al., 2008; MacKinnon and Webb, 2017). Every coefficient is therefore also tested with the wild cluster restricted bootstrap, which imposes the null hypothesis, using Webb's six-point weights (Webb, 2023) because two-point weights give only 2^{12} distinct draws with 12 clusters (9,999 draws, fixed seed). The bootstrap p-value is the one reported in the text.

4.6 Threats to validity

Table 4 lists the main threats, how the design addresses each, and what remains.

5 Data

Servers. All three servers were queried on 26–27 October 2026 through their public MCP endpoints. The *UNICEF SDMX warehouse MCP* (sdmx-mcp 1.2.214) reads the warehouse's public SDMX REST API (`https://sdmx.data.unicef.org`), which holds UNICEF's 41 dataflows, through 31 tools for discovery, structure and data. The *UN System Data Commons MCP* serves a knowledge graph of UN-system statistics, including UNICEF series taken from the SDG Global Database and from UNICEF directly, through six tools and an instruction file it asks agents to read first. The *World Bank Data360 MCP* serves World Bank databases (among them WDI, Health, Nutrition and Population, and the Human Capital Project), which republish

Table 4: Threats to validity

Threat	Type	How the design addresses it	What remains
Agents are not deterministic	internal	fixed prompt; repeat of one setting used as a placebo	one repeat only
Few clusters	statistical	wild cluster restricted bootstrap with Webb weights; sign tests as a check	12 clusters in the model design
Servers change during collection	internal	all collection on 26–27 October 2026; identical calls returned identical content in 67 of 68 cases	results are a snapshot
Different time windows inflate cost	internal	same 2000–2025 window; Data360 re-run	–
The warehouse is also the reference	internal	matched panel; agreement analysis in the sector design	a value the warehouse has wrong is not detected
Scoring involves judgement	construct	automated rules validated against hand scores; element-level status reported	borderline partial scores
Characters stand in for cost	construct	the agentic design measures tokens directly	tokenisers differ across models
The share weights elements equally	construct	every element's status is published, so other weights can be applied	–
The client's own overhead	external	tokens decomposed; only tool definitions and responses attributed to servers	other clients have other baselines
One model family, Nigeria-centred panel, UNICEF indicators	external	17 sectors and three countries in the sector design	other providers, domains and open-ended questions untested

many UNICEF series under World Bank codes, through 16 tools, with data and metadata in separate calls.

Pilot. Two questions for Nigeria, 2000–2025: the under-five mortality rate (UN IGME) and DTP3 coverage (WUENIC), each answered by every reasonable path on each server, with and without a separate metadata call. Data360 was first queried from 1960 and 1980, which made its responses 2.3 and 1.6 times longer, so it was re-run on the same window; both runs are kept and the first is flagged.

Sector and country sample. The frame is the 210 UNICEF indicator codes in UN DC whose series maps to a warehouse series with Nigerian data. Codes are stratified by UN DC's 17 UNICEF sector groups and, within each, ranked by their number of UN DC variables and split into lower and upper halves. One code is drawn per stratum with a fixed seed (20261027), giving 34 series. All are asked for Nigeria, and twelve drawn at random also for Brazil and data-sparse Somalia: 58 questions on the warehouse and UN DC, and 24 on Data360 for the 14 series it carries. Each server follows one path fixed in advance; discovery uses the custodian's indicator name, with one retry using UN DC's name; the data call always uses the target identifier, so retrieval cost is measured even when discovery fails. Protocol and sample were saved before the first call; agents executed 325 calls used in the analysis, and every response is saved verbatim.

Model and effort panel. Four questions for Nigeria, 2000–2025, chosen because all three servers hold the series and its values agree with the warehouse: the under-five mortality rate (UN IGME), the maternal mortality ratio (MMEIG), the population using at least basic drinking water services (JMP), and women aged 15–49 with four or more antenatal care visits (UNICEF household-survey database). The warehouse values are the reference. Each agent received the same prompt, could use only one server (no web search, files or prior knowledge), and had to answer in a fixed JSON form, returning null for anything the server did not give it. Seven settings were run — Claude Haiku 5.5 at medium effort, and Claude Sonnet 5.5 and Claude Opus 5.5 at low, medium and high effort — plus a repeat of Opus at medium effort: 96 sessions. Claude Fable 5.1 was planned, but all 12 of its sessions were refused because the account had no usage credits for it.

6 Results

6.1 How far the servers meet the standard

The warehouse MCP meets 5 of the 15 elements and partly meets 8; Data360 meets 1 and partly meets 9, mostly through its separate metadata call; UN DC meets 0 and partly meets 4 (Figure 1, Table 5). The ranking follows what each server passes through: the warehouse publishes most of what the standard needs, unevenly across its 41 flows; Data360 holds rich series metadata but keeps it apart from the data; UN DC drops every observation-level attribute, so even where it copies warehouse values it loses the bounds, status, sources and notes attached to them. No server meets the retrieval stamp (R2) or licence with citation (S7), the two elements any server can add without touching its data.

6.2 The pilot frontier

On the two pilot questions the cheapest answer delivers between a quarter and two-thirds of the standard, and a compliant answer would cost about the same (Figure 2). For under-five mortality, the cheapest answer is Data360's data call: 7,937 characters and 27% of the standard. For DTP3, the warehouse's two-call answer (7,350 characters, 67%) is on the frontier. Adding its metadata call lifts Data360 to 46–50% at a cost of 16–19 thousand extra characters, more

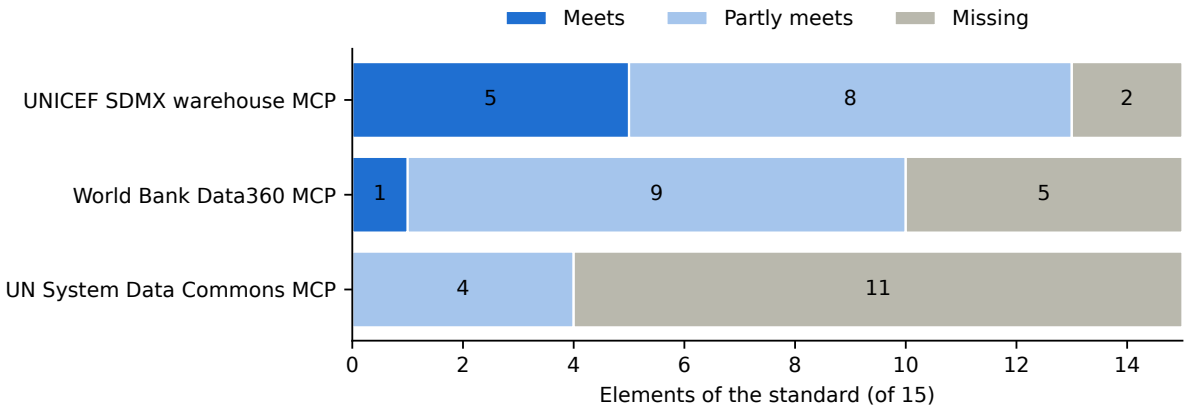


Figure 1: Elements of the standard met in a data response, by server. Meets: present in the response to a data query; partly: incomplete, uneven across flows, or only via a separate call.

Table 5: Element-by-element status (M = meets, P = partly meets, – = missing), with the evidence for the warehouse

Element	Warehouse	UN DC	Data360	Warehouse evidence
R1 Reproducible query	M	P	P	echoes queryURL with flow, key and period
R2 Retrieval stamp	–	–	–	no timestamp or server version in responses
R3 Completeness statement	M	–	P	appliedDefaults, maxObs and the padded key are returned
S1 Identifiers	P	P	P	custodian code and flow key; no SDG series link
S2 Name and definition	P	–	P	definition only from the discovery call; no methodology link
S3 Unit	M	P	M	UNIT_MEASURE code and UNIT_MULTIPLIER; labels on request
S4 Breakdown	M	P	P	full SDMX key with _T totals
S5 Custodian and provenance chain	P	–	P	DATA_SOURCE mixes custodian, survey and pipe; CUSTODIAN only in nutrition
S6 Release and currency	P	–	P	release named inside DATA_SOURCE text for some flows; no date or series status
S7 Licence and citation	–	–	P	no licence or citation in the payload
O1 Source	P	–	P	DATA_SOURCE on 59% of observations; MNCH as bare codes
O2 Reference period	P	–	–	COVERAGE_TIME filled in six flows, nutrition and HIV among them (9% of observations); not in MNCH
O3 Nature and status	P	–	–	OBS_STATUS on 49%; none in WASH, gender, education, economic flows
O4 Uncertainty	P	–	–	bounds on 52% of observations; interval level never stated
O5 Footnote	M	–	–	observation, series and country notes passed through

than half of them lists of countries, regions and years; UN DC stays at 19–21% whether or not it adds its own metadata call.

A standard-compliant response, built from the warehouse's own values and attributes with unpublished fields left as placeholders, takes 2,834 characters for 25 years of under-five mortality (2,929 once the placeholders are filled at typical lengths), fewer than the warehouse's current data response. With the warehouse's discovery call, it costs 9,149 and 4,747 characters for the two questions: about 15% more than the cheapest answer measured today for under-five mortality, and less than any answer measured for DTP3, for 100% of the standard.

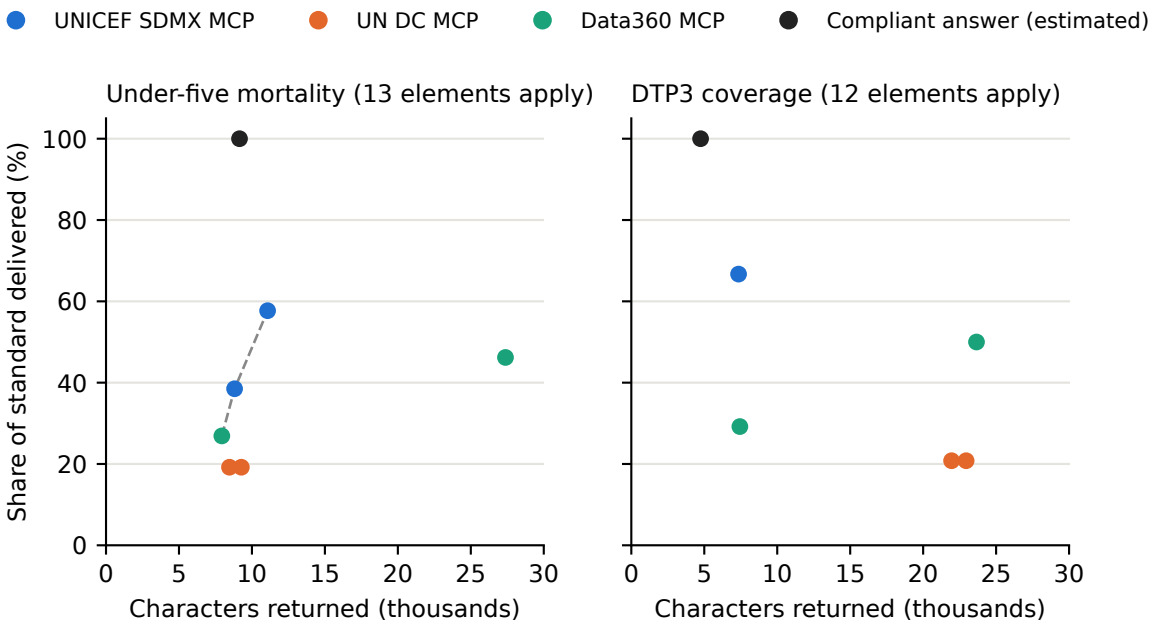


Figure 2: The pilot frontier: characters returned against share of the standard delivered, Nigeria 2000–2025. Dashed line: measured frontier.

6.3 Across sectors and countries

The pre-registered benchmark confirms the ranking on the standard but overturns the ranking on cost (Figure 3). On all 51 questions that both servers answered, the warehouse delivered more of the standard than UN DC: 46–70% against at most 21%. Data360 delivered 20–29%. But UN DC's answer was cheaper than the warehouse's on 43 of 58 questions. Median characters per question were 13.4 thousand for the warehouse, 8.9 thousand for UN DC and 4.2 thousand for Data360. The pilot questions had flattered the warehouse: its search returns a median 9.6 thousand characters for a custodian indicator name.

Search, not metadata, is what costs (Figure 4). Discovery takes 94% of UN DC's characters and 67% of the warehouse's, plus 10% for the warehouse's dimension listing. The data call, which carries the metadata, has a median of 1.9 thousand characters on the warehouse and 0.6 thousand on UN DC. UN DC's search has a long tail, up to 128 thousand characters for migrants in Brazil. Searches cost more for indicators with more breakdowns (Spearman correlation with the number of UN DC variables per code: 0.36 on UN DC, 0.60 on the warehouse). Data360 is cheap partly because its search fails: it found 4 of the 14 series it carries.

All three servers answered every Nigerian question. The warehouse and UN DC each answered 10 of 12 questions for Brazil and 7 of 12 for Somalia, missing the same ones because UN DC copies the warehouse's gaps. A question with no data costs the warehouse a median 12.6 thousand characters and UN DC 0.6 thousand, because UN DC's search filters by country first.

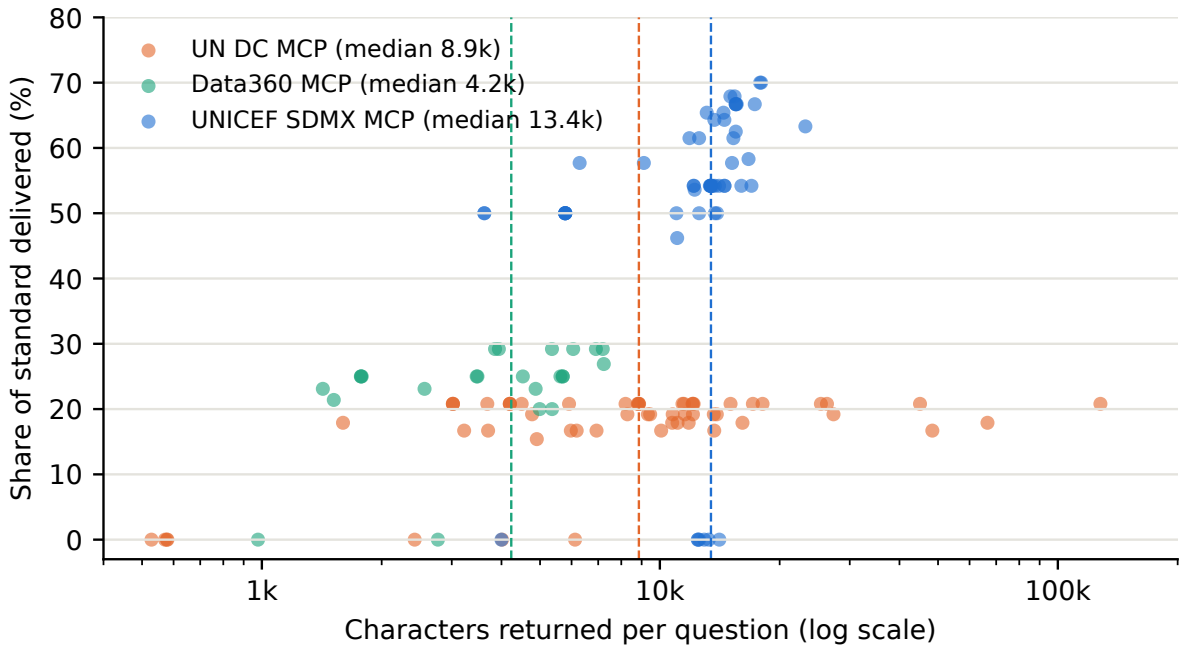


Figure 3: Characters and share of the standard for 140 answers to 58 questions (34 series; Nigeria, Brazil, Somalia). Unanswered questions at 0%; dashed lines are median characters per server.

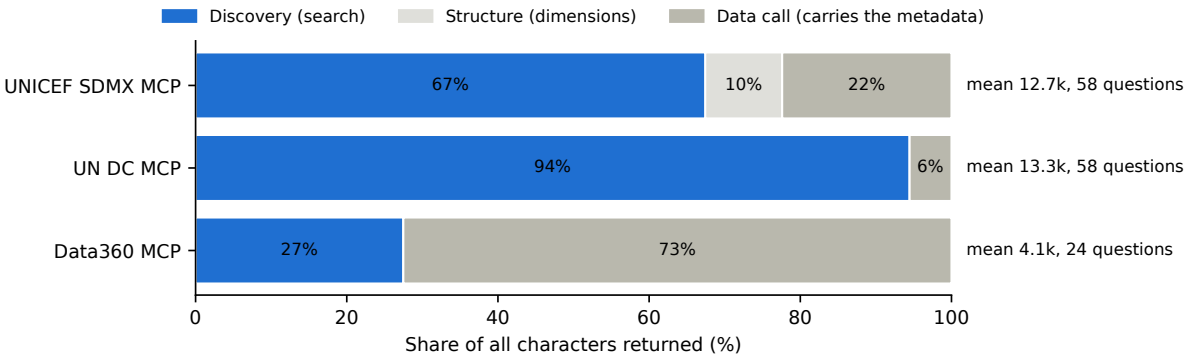


Figure 4: Share of all characters returned, by component of the answer; sector and country benchmark, fixed paths.

UN DC's values matched the warehouse's in every common year on 41 of 51 answered questions; most of the rest are older releases. Its latest year was the same on 37, earlier on 11 and later on 3. Data360 matched fully on 12 of 19 questions with common years, and its latest year was earlier on 6 of 22. Table 6 gives the medians by sector.

Table 6: Median characters and share of the standard, by sector (unanswered questions included)

Sector	Questions	UNICEF SDMX MCP	UN DC MCP	Data360 MCP
Child mortality	2	7.7k ·58%	9.4k ·17%	4.9k ·23%
Demography	2	16.1k ·60%	5.3k ·19%	7.2k ·29%
Early childhood development	2	15.2k ·68%	11.4k ·18%	–
Economy and aid	6	5.8k ·50%	3.6k ·21%	–
Education	2	13.8k ·52%	10.9k ·20%	4.5k ·25%
Functional difficulty	4	13.8k ·52%	17.4k ·19%	–
Gender	4	12.4k ·52%	6.7k ·21%	–
HIV/AIDS	4	17.6k ·68%	4.7k ·17%	–
Immunization	4	15.5k ·67%	11.7k ·21%	1.4k ·23%
Migration	6	13.8k ·54%	19.2k ·21%	2.9k ·27%
Maternal, newborn and child health	2	15.2k ·60%	11.8k ·19%	4.9k ·25%
Nutrition	2	19.4k ·65%	31.0k ·17%	5.2k ·20%
Child protection	2	14.1k ·64%	13.4k ·18%	1.5k ·21%
Child poverty	4	12.3k ·25%	1.1k ·9%	–
Social policy	4	3.8k ·48%	4.4k ·17%	3.9k ·29%
WASH	6	13.4k ·54%	8.8k ·21%	4.6k ·25%
Adolescents	2	13.5k ·63%	10.1k ·19%	–

6.4 Does the model or the effort level matter?

Every setting answered all 12 questions, and every value reported for a year the warehouse holds was correct (Figure 5, Table 7). Of the 1,953 year-values reported across the 96 sessions, all 1,952 for years the warehouse holds match it within rounding; the other is a year (2000) from a Data360 series the warehouse does not hold. Every reported number, including all 290 uncertainty bounds, appears in the server's own responses. The reference has 246 year-values per setting; the two missing in every row are a coverage gap in Data360's antenatal care series.

Table 7: Cost and accuracy by model and effort setting (12 questions each)

Setting	Tokens sent per question	Tokens received per question	MCP calls	Characters per question	Years correct / reported	Metadata fields
Haiku, medium	655k	2,860	62	27.0k	244 / 244	7.9
Sonnet, low	587k	1,483	47	19.3k	244 / 244	7.9
Sonnet, medium	588k	1,569	46	18.0k	244 / 244	8.2
Sonnet, high	643k	1,805	57	21.9k	244 / 244	8.4
Opus, low	551k	1,308	38	22.5k	244 / 245	7.5
Opus, medium	639k	1,631	51	20.9k	244 / 244	8.1
Opus, high	690k	1,823	57	20.4k	244 / 244	8.1
Opus, medium (repeat)	563k	1,608	47	20.0k	244 / 244	8.1

Medians per question, except MCP calls (total over 12 questions) and metadata fields (mean).

Completeness depended on which series the agent picked. Data360 holds antenatal care in two series: the Human Capital Project series starts in 2010 and misses 2003 and 2008; the Health, Nutrition and Population series ends in 2018 and misses the two latest surveys. Only Opus at low effort picked the second, so its latest value was from 2018.

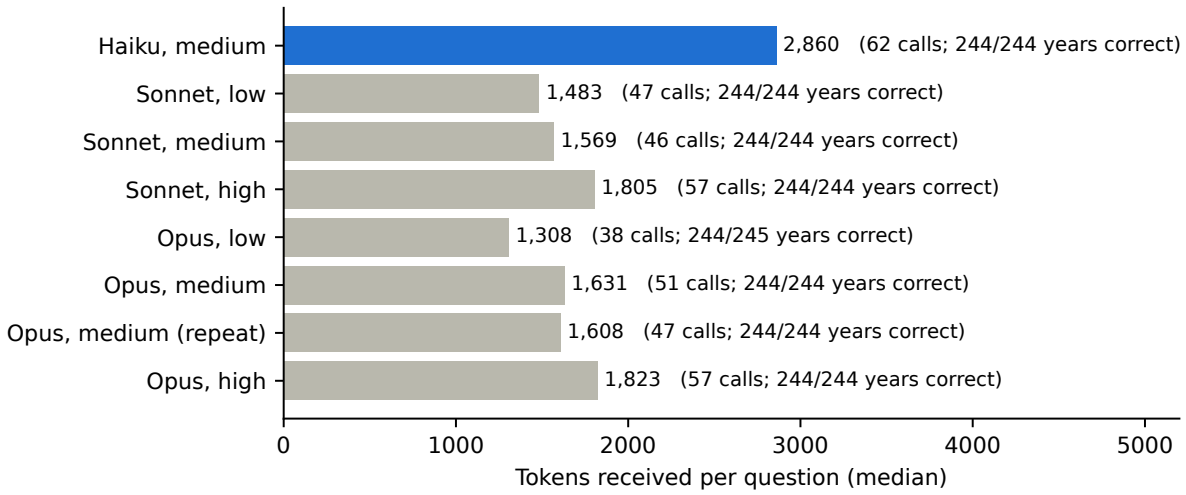


Figure 5: Tokens received per question (median), MCP calls and years correct, by model and effort setting; 12 questions per setting.

Higher effort bought more calls, not better answers. For Opus, MCP calls rose from 38 to 51 to 57 across low, medium and high effort, and tokens received from 1,308 to 1,631 to 1,823 per question. Sonnet's tokens received rose in the same way (1,483 to 1,805). Haiku made the most calls (62) and generated the most tokens: 2,860 per question, about 1.8 times Sonnet at medium effort. Section 6.5 estimates these differences with cell fixed effects and tests them against a placebo.

The values were stable; the cost and the metadata were not (Table 8). Repeating Opus at medium effort reproduced the series and values in all 12 cells. Tokens sent changed by a median of 2% per cell and at most 22%, much less than the effect of effort compared within the same cells (median +33% for Opus; Section 4). The servers behaved deterministically: on UN DC, 11 of 18 repeated calls were byte-different only because JSON keys came back in a different order. One warehouse call, `get_flow_structure` for the child-mortality flow, returned 1.3 million characters; the client refused it and the agent answered without it.

Table 8: Consistency and grounding

Check	Result
Question × server cells where all 8 runs returned identical values	11 of 12
Cells where all 8 runs chose the same series identifier	9 of 12
Largest spread in metadata fields reported within a cell	5 fields
Repeat run: same series and values	12 of 12 cells
Repeat run: same number of MCP calls	9 of 12 cells
Identical calls in different sessions returning identical content	67 of 68
Reported values found in the server's responses	1,953 of 1,953
Reported uncertainty bounds found in the server's responses	290 of 290

Where the tokens go. Tokens sent were about 356 times tokens received. Most are not caused by the server: 84% is the client's own baseline of about 97,000 tokens re-sent with every request, and 88% of all tokens sent were cache reads. Of what the server adds, its tool definitions dominate (Figure 6). Loading a server's tools adds about 15,000 tokens (UN DC, six tools), 20,000 (warehouse, 31 tools) or 22,000 (Data360, 16 tools) to every later request. Over a question of 4 to 6 requests that is 70,000–80,000 tokens on average, against 17,000–33,000 for all responses and reasoning together. Tool count alone does not set the cost: UN DC's six tools

weigh almost as much as the warehouse's 31, because their descriptions are long. By characters, the largest items were Data360's `get_metadata` (18,600 per call) and UN DC's instruction file (12,000, read once per session); without it, UN DC needed a median of 8.1 thousand characters per question, against 16.6 thousand for the warehouse and 27.3 thousand for Data360.

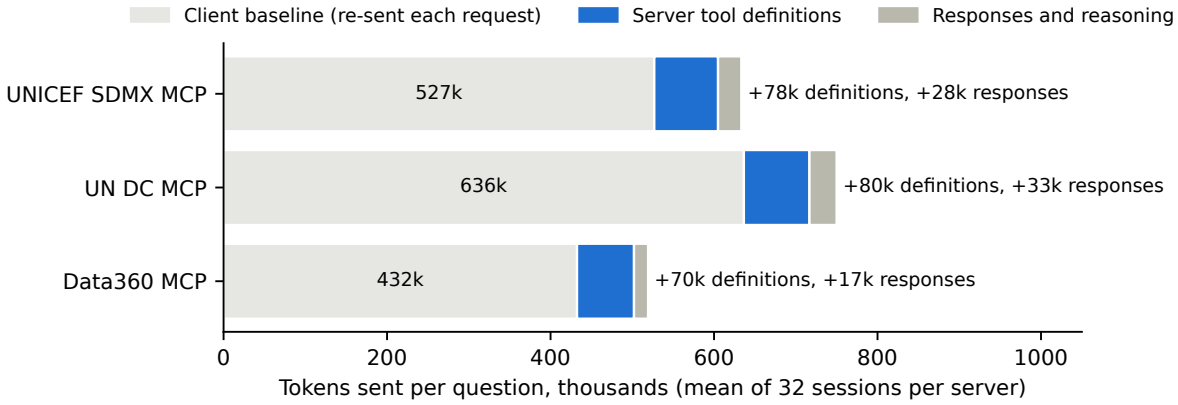


Figure 6: Tokens sent per question by component, mean of 32 sessions per server. Definitions: the jump in context when the server's tools load, times later requests.

6.5 Average effects

Table 9 reports the estimates of equations (1) and (2). Within the same question, UN DC delivers 37.9 percentage points less of the standard than the warehouse and Data360 31.6 points less (bootstrap $p < 0.001$ for both). UN DC's answers are 35.2% cheaper in characters than the warehouse's on average ($p = 0.007$) and Data360's 75.7% cheaper ($p < 0.001$), although the latter reflects searches that failed.

High effort raised tokens sent by 36.5% for Opus ($p < 0.001$) and 12.6% for Sonnet ($p = 0.003$), and tokens received by 38.8% and 22.0%. Haiku received 77.4% more tokens than Sonnet at medium effort ($p < 0.001$), but sent only 11.5% more ($p = 0.051$). The placebo is informative: the repeat of Opus at medium effort sent 7.3% fewer tokens than the first run ($p = 0.011$), although nothing but chance separates them. Average differences of about ten per cent in tokens sent are therefore within run-to-run variation: of the effects on tokens sent, only that of Opus effort clearly exceeds it, while Sonnet's effort (+12.6%) and Haiku (+11.5%) do not. On tokens received the placebo is smaller (5.4% fewer, $p = 0.052$), and the effects of effort for both models and of Haiku all exceed it. Clustering matters in the server comparisons, where cluster-robust standard errors are up to 86% larger than heteroskedasticity-robust ones, but hardly in the model comparisons, where the cell fixed effects absorb most of what sessions in a cell share. Bootstrap p-values are smaller than CR1 ones for 19 of the 20 estimates, and only 1 conclusion changes at the 5% level: Haiku's effect on MCP calls, significant with the bootstrap ($p = 0.015$) but not with CR1 ($p = 0.063$).

7 Policy implications

7.1 For an official UNICEF MCP

1. **Return values and the fifteen elements in one data call.** The metadata travels in the cheap part of the answer, and an agent rarely makes a second call. The gaps to close are a retrieval stamp (R2), a structured release and currency field (S6), licence and citation (S7), and status and bounds in the flows where they are empty (O3, O4), with the interval level stated.

Table 9: Average effects: fixed-effects estimates with cluster-robust inference

Contrast	Cluster (G)	Estimate	%	SE		p	
				CR1	HC1	CR1	WCR
A. Servers, equation (1): difference from the warehouse, question fixed effects							
Share of the standard (points), answered questions							
Data360	series (34)	−31.640		2.138	1.285	< 0.001	< 0.001
Data360	question (51)	−31.640		1.688	1.285	< 0.001	< 0.001
UN DC	series (34)	−37.922		1.891	1.018	< 0.001	< 0.001
UN DC	question (51)	−37.922		1.400	1.018	< 0.001	< 0.001
ln characters, all questions							
Data360	series (34)	−1.414	−76%	0.350	0.190	< 0.001	< 0.001
Data360	question (58)	−1.414	−76%	0.238	0.190	< 0.001	< 0.001
UN DC	series (34)	−0.435	−35%	0.189	0.134	0.028	0.007
UN DC	question (58)	−0.435	−35%	0.179	0.134	0.018	0.001
B. Model and effort, equation (2): contrasts between settings, cell fixed effects, $n = 96$							
ln tokens sent							
Opus: high vs low effort	cell (12)	+0.311	+36%	0.069	0.065	< 0.001	< 0.001
Sonnet: high vs low effort	cell (12)	+0.119	+13%	0.047	0.046	0.027	0.003
Haiku vs Sonnet (medium)	cell (12)	+0.109	+12%	0.056	0.057	0.076	0.051
Repeat of Opus medium (placebo)	cell (12)	−0.076	−7%	0.033	0.046	0.041	0.011
ln tokens received							
Opus: high vs low effort	cell (12)	+0.328	+39%	0.055	0.060	< 0.001	< 0.001
Sonnet: high vs low effort	cell (12)	+0.199	+22%	0.047	0.040	0.001	< 0.001
Haiku vs Sonnet (medium)	cell (12)	+0.573	+77%	0.071	0.064	< 0.001	< 0.001
Repeat of Opus medium (placebo)	cell (12)	−0.055	−5%	0.029	0.042	0.080	0.052
MCP calls							
Opus: high vs low effort	cell (12)	+1.583		0.442	0.460	0.004	< 0.001
Sonnet: high vs low effort	cell (12)	+0.833		0.301	0.296	0.018	0.006
Haiku vs Sonnet (medium)	cell (12)	+1.333		0.646	0.541	0.063	0.015
Repeat of Opus medium (placebo)	cell (12)	−0.333		0.209	0.315	0.139	0.106

Notes: OLS. Panel A clusters by series; clustering by question is shown as a sensitivity check. Panel B clusters by question × server cell. % is $100(e^{\beta} - 1)$ for outcomes in logarithms. p (CR1): cluster-robust variance with $t(G - 1)$; p (WCR): wild cluster restricted bootstrap with Webb weights, 9,999 draws.

2. **Make discovery compact.** Search should return a short ranked list of identifiers and names, filtered by country first, and leave definitions, units and flow lists to the data response; this is where 67% of the warehouse's characters go, plus 10% for structure.
3. **Keep the tool list short and its descriptions tight.** Tool definitions are re-sent with every request and in agentic use cost more tokens than the data. A small core set with advanced tools behind it would cut the largest server-attributable cost.
4. **Cap and page every response**, so that no single call can return 1.3 million characters.
5. **Declare the response schema** through `outputSchema` and return structured content that conforms to it.
6. **Test conformance and cost continuously**, with the fixed query sets and scoring scripts used here, reporting share of the standard, characters, tokens sent and received, accuracy and stability across repeated runs.

7.2 For republishers

Pass through, never drop: almost all of UN DC's gaps are attributes the warehouse already publishes. Start with custodian attribution (S5) and release (S6), which answer the attribution concerns custodian agencies have raised and expose stale copies. Mark the canonical series: agents split between duplicate copies of the same series, and where the copies differ in coverage the choice changed the answer.

7.3 For the statistical community

The standard reuses SDMX and SDG attributes, so it asks custodians for nothing new. It can be tabled at the CCS-UN as the minimum any server that republishes custodian data should meet, with S5, S6 and S7 first, and proposed to the UN System Data Commons under UN80 Workstream 16.

7.4 For evaluating agents on official data

Monitor tokens sent and received separately: they differ by two orders of magnitude and respond to different design choices. Do not assume a smaller model is cheaper per answer, or that higher effort improves accuracy on well-specified questions. Estimate the effect of a setting with question $\times$ server fixed effects, cluster standard errors by that cell, and use the wild cluster bootstrap when cells are few. Include an identical repeat as a placebo: here it differed from the first run by 7.3% in tokens sent, so effects of that size cannot be told from noise with one run per setting.

8 Final remarks

A modest standard of fifteen elements, built entirely from attributes the statistical system already defines, would let an agent reproduce, attribute, date and interpret every value, at no more than the cost of today's cheapest answers. The servers differ far more in what they pass through than in what they cost; cost is driven by search and by the tool list rather than by the data; and the model or effort level changes the cost but not the values. The fixed-effects estimates have the same sign as the descriptive and paired results on every comparison. The design of the server, not the choice of model, decides whether an answer is complete and attributable.

Limitations. The agentic benchmark covers four series for one country with one repeat; the sector benchmark covers three countries; both are snapshots of servers that change. The warehouse is both a server under test and the reference, so a value it has wrong would not be

detected. The client's baseline is specific to the agent harness used, although the tool-definition and response components are not. Claude Fable 5.1 and other model families were not tested. The prompt was well specified; open-ended or ambiguous questions may separate models more. Partial scores involve judgement, although the automated rules reproduce the hand scores of the pilot. With one run per setting and 12 clusters, the model-and-effort design can reliably detect effects on tokens sent of about 15–24% or more (two-sided 5% test, 80% power); smaller effects, such as those of Sonnet's effort and of Haiku on tokens sent, are not resolved.

Next steps. Run the agentic benchmark with at least three repeats per setting, which would shrink the placebo and the detectable effect, on more countries and open-ended questions; run all three benchmarks against the prototype of the official UNICEF MCP before and after implementing the standard and a compact search; and publish the query sets and scoring as a conformance test any republishing server can run.

Reproducibility

Every result in the text is a macro written by `paper/code/assets.py` from the analysis outputs (only design constants, such as sample sizes, dates and tool counts, and the earlier comparisons cited in Section 2 are written out), every table is generated and every figure is drawn from the same files; the fixed-effects models, clustered standard errors and bootstrap are computed by `analysis/effects.py`; `make all` rebuilds the analysis, the assets and this PDF from the saved server responses and agent transcripts in the repository. Collection (querying live servers with agents) is documented but not re-executed by `make`, since the servers change.

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